Sentiment Analysis on Cinematic Reviews: Fine-Tuning BERT for Binary Text Classification

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1. Problem Formulation and Dataset Description

The objective of this project is to implement, fine-tune, and evaluate a transformer-based Large Language Model for binary text classification. Specifically, the model is trained to accurately determine the underlying sentiment (positive or negative) of long-form, unstructured movie reviews. This task presents unique challenges, as cinematic reviews are often lengthy, utilize complex or sarcastic language, and lack a rigid grammatical structure. By reproducing an open-source NLP workflow, this project demonstrates a complete end-to-end pipeline—from data preprocessing and tokenization to model fine-tuning and deployment.

Dataset Description

This implementation utilizes the IMDB Dataset, a benchmark NLP dataset containing 50,000 highly polar movie reviews. The dataset is evenly split into 25,000 reviews for training and 25,000 for testing, ensuring a robust evaluation phase. The labels are perfectly balanced, corresponding to negative (0) and positive (1) sentiments, which prevents the model from achieving artificially high accuracy simply by guessing the majority class.
To efficiently process the data, the Hugging Face datasets library was utilized to load, batch, and map the text data into PyTorch tensors. To optimize computational time during the evaluation phase, the final model performance was validated on a random 5,000-review subset of the test split, providing a reliable estimate of generalization performance without requiring a full evaluation loop over the 25,000 items.

2. Model Architecture and Training Details

Architecture

The core of the classification pipeline relies on bert-base-uncased (Bidirectional Encoder Representations from Transformers). This foundational model features 12 layers, a hidden size of 768, 12 attention heads, and approximately 110 million parameters. Unlike traditional models that read text sequentially left-to-right, BERT processes tokens bidirectionally, allowing it to establish rich contextual dependencies from both directions surrounding a given word.
To adapt the model for classification, a randomly initialized sequence classification head (a linear layer) was appended on top of the pooled output of the pre-trained BERT model. During training, the base transformer weights were fine-tuned concurrently with this new classification head to aggressively optimize for the binary classification parameters of the IMDB dataset.

Data Processing and Training Logistics

Raw review text was tokenized using the BERT tokenizer. To manage GPU memory constraints and ensure training efficiency, all text inputs were strictly truncated to a maximum length of 256 tokens. The pipeline also deployed dynamic padding via a data collator, ensuring that each batch was only padded to the length of the longest review within that specific batch, saving vast amounts of memory overhead.

Hyperparameter	**Value**	**Rationale**
Epochs	2	Sufficient to converge without inducing extreme overfitting on the IMDB training corpus.
Batch Size	16	Optimal balance between training speed and the hardware limits of the free Google Colab T4 GPU.
Learning Rate	2e-5	A small learning rate prevents the catastrophic forgetting of established pre-trained base weights.
Precision	fp16	Mixed precision accelerates

Hyperparameter	Value	Rationale
		tensor calculations significantly on the selected GPU hardware.

3. Evaluation Metrics and Analysis

The model was evaluated using accuracy and the F1-Score (calculated via the scikit-learn library). Accuracy tracks the overall percentage of correctly classified reviews, while the F1-Score operates as the harmonic mean of precision and recall. Utilizing the F1-Score ensures the model is penalized if it disproportionately favors either the positive or negative class.

- **Epoch 1:** 90.6% Accuracy | 0.911 F1-Score
- **Epoch 2:** 92.1% Accuracy | 0.922 F1-Score

Across the training duration, the training loss fell significantly from approximately 0.25 to 0.16. In contrast, the validation loss remained relatively flat, shifting only marginally from 0.276 to 0.281. This specific divergence on the loss curve is indicative of mild overfitting; the model began to heavily memorize the intricacies of the training data rather than generalizing its feature recognition to unseen data. However, because overall accuracy still increased between the first and second epoch, halting training at two epochs proved to be the optimal stopping point.

4. Reflection on Encountered Errors or Difficulties

Throughout the development and deployment lifecycle of this NLP pipeline, several distinct technical challenges were encountered and subsequently resolved:

- **Dataset Loading Error (HfUriError):** During the initial configuration, attempting to load the IMDB dataset triggered an HfUriError. Hugging Face's modernized API requires repository IDs to follow a specific namespace format. This was swiftly debugged by altering the call from the generic imdb alias to the fully qualified stanfordnlp/imdb namespace.
- **Lost Model Weights via Reset:** A major setback occurred when the Google Colab session disconnected due to an idle timeout, which entirely wiped the runtime environment and deleted the fine-tuned model weights. This missing model data subsequently caused the web application to throw a "Could not parse server response" error. To permanently resolve this, the model was retrained, and the pipeline was updated to save the final weights directory to a permanently mounted Google Drive.

- **Variable Tensor Sizes in Batching:** Movie reviews exhibit extreme variance in length, which natively causes PyTorch tensor size mismatches when batching data during the collate phase. This was mitigated by implementing strict truncation at 256 tokens and applying dynamic padding, enabling every batch to cleanly match the dimensionality of its internal longest sequence without excessive padding overhead.